

Functional Features Included in the Solution

OptiCrop — Smart Agricultural Production Optimization Engine

Implemented Features (3 core pages)

- 1. Home page — project introduction and navigation entry point.
- 2. About page — explains the project's purpose, methodology, and model performance.
- 3. Find Your Crop page — interactive form collecting N, P, K, temperature, humidity, pH, rainfall, returning a real-time crop recommendation from the trained model.

Supporting ML Features

- Multi-algorithm comparison (Logistic Regression, KNN, Decision Tree, Random Forest) with Accuracy, Precision, Recall, and F1-Score reported for each.
- K-Means clustering for exploratory grouping of soil/climate patterns.
- Outlier detection and correction (IQR method + log transform) on the Potassium feature.
- Derived seasonal classification feature (Winter / Summer / Rainy / Moderate) based on each crop's average climate profile.

Total Feature Count

3 user-facing pages/features + 4 supporting ML/analysis features = 7 functional components, all implemented and verified working.